Design Research Society
DRS Digital Library

Textile Intersections Conference Series | Textile Intersections 2026

Bacterial Cellulose as Biofabricated Textile Material: Biotinkering and Trajectories of Textile Materialisation

Ziqian Yu
Politecnico di Milano, Italy

Valentina Rognoli
Politecnico di Milano, Italy

Follow this and additional works at: https://dl.designresearchsociety.org/textileintersections

 Part of the Arts and Humanities Commons

Citation

Yu, Z.,and Rognoli, V.(2026) Bacterial Cellulose as Biofabricated Textile Material: Biotinkering and Trajectories of Textile Materialisation, in Jane Scott, Sophie Skach, and Rebecca Stewart (eds.), *Textile Intersections 2026*, 17-18 September 2026, London, United Kingdom. https://doi.org/10.21606/TI-2026.179

This Research Paper is brought to you for free and open access by the DRS Special Interest Groups at DRS Digital Library. It has been accepted for inclusion in Textile Intersections Conference Series by an authorized administrator of DRS Digital Library. For more information, please contact library@thedrs.org.

Design Research Society DRS

TEXTILE INTERSECTIONS

Dyson School of Design Engineering, Imperial College London
16 – 18 September 2026

Bacterial Cellulose as Biofabricated Textile Material: Biotinkering and Trajectories of Textile Materialisation

Yu, Ziqian*[a]; Rognoli, Valentina[a]

[a] Politecnico di Milano, Milan, Italy
* ziqian.yu@polimi.it

Bacterial cellulose (BC) is increasingly studied in material design and biodesign as a biofabricated material shaped through cultivation, harvesting, and processing. In fashion contexts, however, BC is often identified as a leather alternative, particularly in sheet form. While this framing has supported its integration into design practices, it constrains BC's textile potential. This paper investigates how BC becomes textile material through multiple materialisation trajectories rather than through a single substitute material form. The study is a comparative analysis of sixteen cases, framed by Material-Driven Design (MDD) approach. Fourteen external design and interdisciplinary cases provide the comparative basis, while two hands-on experiments by the authors, developed through Research through Design (RtD) and biotinkering, examine BC as a natural binder for textile waste and within a BC–alginate extrusion system for filaments. Through this analysis, six trajectories of textile materialisation are identified: sheet, fibre, filament, yarn, biocomposite textile, and textile coating. These trajectories are shaped by different relations between cultivation, processing, textile assembly, and surface treatment, showing that BC's textile potential does not reside in one stable form but emerges through how biological growth and material transformation are distributed across textile practices. The paper argues that BC should be understood as a process-based biofabricated textile material rather than a leather substitute, offering a more grounded basis for engaging with BC in textile design, biodesign, and emerging material practices.

Keywords: bacterial cellulose; biofabricated textile material; biotinkering; materialisation trajectories

Publisher: DRS (© The Author(s)). The authors of this paper acknowledge the right of Design Research Society to publish the paper under the Creative Commons Attribution 4.0 International (CC BY-NC 4.0) licence.
Rights: This work is made available according to the conditions of the Creative Commons Attribution 4.0 International (CC BY-NC 4.0) licence. Full details of this licence are available at: https://creativecommons.org/licenses/by-nc/4.0/.
DOI: 10.21606/TI-2026.179

©2026

1. Introduction

Bacterial cellulose (BC), a biofabricated material produced through microbial fermentation, has gained attention in biodesign and textile research (Wang et al., 2019; Camere & Karana, 2018; Nayak et al., 2024). Much of this interest defines BC as a sustainable material and an alternative to animal leather (Absharina et al., 2025; Provin et al., 2021; Wood et al., 2023). This paper takes that motivation as the starting point, but the concern is not whether BC is sustainable, it is how BC becomes a textile material. Environmental sustainability enters only where it bears on this question: at which points in the materialisation process environmental and processing interventions are located.

In textile and fashion contexts, this alternative is most often approached through BC's sheet form, a framing this paper examines and moves beyond. Less attention has been given to how BC becomes textile material through fibre making, filament formation, yarn construction, scaffold use, textile waste binding, or coating (Morrow et al., 2023; Harmon, 2019; Takagi et al., 2025; Ogrizek et al., 2024). Studies of these forms and of design-led biofabrication projects remain dispersed across material science, fashion, biodesign, and textile design (Roussel et al., 2024; Nayak et al., 2024; Fernandes, Silva & Silva, 2025). A textile design account is needed to connect these practices.

This paper asks: How can bacterial cellulose be understood as a biofabricated textile material through different trajectories of textile materialisation? More specifically, it asks how BC becomes textile material through growth, processing, textile assembly, and surface treatment. To address this question, the paper examines sixteen cases, comprising two authors' experiments and fourteen external design and interdisciplinary cases. Through comparative analysis, it identifies six trajectories of textile materialisation: sheet, fibre, filament, yarn, biocomposite textile, and textile coating. The paper argues that BC's textile potential lies not in a single final form, but in how it is grown and processed within textile systems.

2. Background

2.1. Bacterial cellulose as an emerging material in design

Bacterial cellulose is produced by acetic-acid bacteria such as *Komagataeibacter* and *Acetobacter* species, which secrete cellulose nanofibrils at the air–liquid interface of a sugar-rich medium and form a hydrated pellicle (Wang et al., 2019; Absharina, Putra, et al., 2025). In design contexts, BC is associated with kombucha cultures and other small-scale fermentation settings (Roussel et al., 2024; Verran et al., 2023). It can be cultivated without heavy industrial infrastructure, making it accessible for material exploration in biodesign and design-led biofabrication (Camere & Karana, 2018; Cohen et al., 2023; Cleveland et al., 2024). Related research also explores agro-industrial and food by-products as feedstocks for BC production, although these studies are not textile-specific (Avci et al., 2026; Miranda et al., 2026).

Design research has studied BC alongside living and grown materials (Karana et al., 2015; Karana et al., 2023). Growth, harvesting, drying, and further treatment shape the material qualities of BC. Medium composition, cultivation time, vessel shape, drying conditions, and finishing methods affect thickness, flexibility, shrinkage, stiffness, surface quality, and stability (Wang et al., 2019; Provin et al., 2021; Sözcü et al., 2026). BC is therefore not simply a bio-based material, but a grown material whose qualities form through material work.

This makes BC relevant to wider work on biofabrication, fashion, material design, and biodesign (Mihaleva, 2021; Obregón, 2026; Collet, 2018; Duarte Poblete, 2024). In these fields, material properties are not fixed data given in advance. They are qualities explored through making, testing, and close material engagement (Karana et al., 2023; Parisi et al., 2017). It has also been discussed in relation to grown textiles (Collet, 2017), open-source biofabrication infrastructures (Cohen et al., 2023), and circular fashion practices (D'Itria et al., 2026). BC is positioned as an emerging material whose properties emerge from the growth environment, harvesting time, and processing interventions (Duarte Poblete et al., 2023).

2.2. Bacterial cellulose in textile research

When BC enters textile design and fashion sector, it is often framed as a sustainable leather alternative (Nayak et al., 2024; Provin et al., 2021; Wood et al., 2023). Studies in this area evaluate BC sheets against conventional leather using textile industry tests and discuss wettability, mechanical fragility when dry, and dimensional instability. Recent work has explored dyeing, printing, and surface treatments (Tang et al., 2025), genetically engineered self-pigmentation (Walker et al., 2025), finishing strategies for dehydrated BC films (Phan & Okubayashi, 2025), and BC adhesion to plasma-treated cotton substrates (Ogrizek et al., 2024). The sheet remains the dominant material-form (Fernandes et al., 2025).

Design interest in BC in fact reaches back over a decade, to early biofabrication work such as Suzanne Lee's BioCouture, which grew cellulose pellicles into leather-like garments and brought grown textiles to wider design and public attention in the early 2010s. Yet despite more than a decade of sustained research and design activity, BC's industrial and commercial uptake in the textile sector has remained limited, and it has not displaced conventional leather or established man-made materials at scale.

On the production side, static fermentation remains comparatively slow, space- and labour-intensive, and difficult to standardise, and BC textiles continue to face the recurring obstacles of wet-state processing, mechanical fragility, dimensional instability, and cost noted above (Absharina, Putra, et al., 2025; Provin et al., 2021; Avci et al., 2026). Measured against conventional leather on industrial metrics, the very comparison invited by the leather-substitute framing, BC therefore tends to appear as an underperforming alternative. Its continued, and indeed growing, research appeal is better explained not by market readiness but by its value as an accessible and generative material for design exploration, open to intervention at many points in its growth and processing, a richness that the dominant sheet-and-substitute framing largely leaves unexamined.

A smaller but expanding body of work has begun to test BC beyond the sheet. BC has been wet-spun and co-designed into filament-like forms (Morrow et al., 2023); high-molecular-weight BC has been doped into regenerated cellulose textile-waste fibres to improve their spinnability (Silva et al., 2024; Moriam et al., 2026). BC has been grown into hierarchical structures with oriented fibrils for pressure-sensing fabrics (Gao et al., 2024); and green-chemistry electrospinning routes for BC have been reported (Vasili et al., 2025). In adjacent textile-bio practices, knitted textiles have hosted scent-producing bacterial species (Nerlich et al., 2023), and growing pigments and patterns have been explored as living textile design (Moser et al., 2023). Work on nanocellulose and bio-based textile materials shows the need to connect material development with textile performance and textile design practice (Biranje et al., 2024; Kapsali & Hall, 2023; Kapsali et al., 2024). These examples are part of textile design discussions on biotextiles, performative textile interfaces, and sustainable textile practices (Buso et al., 2023; Morgan et al., 2024).

However, these developments are not yet well connected from a textile design perspective. Scientific papers often focus on production routes, properties, or performance, while design and commercial cases often show material practices without full technical detail. Mapping work on kombucha BC has shown the diversity of design and fabrication practices around the material (Roussel et al., 2024), but what remains missing is not another general survey of BC applications. The gap is a clearer account of how BC becomes the material through specific textile practices.

2.3. Material-Driven Design, Material-Driven Textile Design and biotinkering

Material-Driven Design frames this study by defining material exploration as a way of producing design knowledge (Karana et al., 2023; Karana et al., 2015). In this view, materials are not only chosen to serve predefined concepts. They guide questions, reveal constraints, and shape design work. Material-Driven Textile Design extends this frame to surface formation, fibre and yarn construction, filament making, coating, scaffold use, and material assembly (Ribul et al., 2021).

Material tinkering describes an open-ended, hands-on engagement with materials in which designers learn through manipulation rather than from datasheets, and through which unexpected material qualities can be discovered (Parisi et al., 2017). It is especially relevant during wet handling, drying, binding, extrusion, and contact with textile-related materials. In this paper, biotinkering describes the practical way in which the authors worked with BC to observe material behaviour and generate situated material knowledge.

Biotinkering is here understood as the extension of material tinkering into biologically mediated making, a mode of design practice that stays close to the wet, the grown, and the unfinished, and that treats the designer–microbe relation as part of the formation process (Bell et al., 2026; Lazaro Vasquez et al., 2026; Yu et al., in press). Biotinkering is not the only possible frame for biofabricated textile design, but it foregrounds the processes through which BC becomes textile rather than the forms it eventually takes.

3. Method

3.1 Research design and case studies

This study employs a qualitative, comparative case analysis: sixteen cases of BC in textile contexts are examined and coded into trajectories of textile materialisation according to how BC is cultivated, processed, assembled, and treated. The two authors' experiments use Research through Design (RtD), enacted through biotinkering, to generate situated material knowledge from hands-on experimentation (Schön, 1983; Stappers, 2014; Karana et al., 2023). The aim is not to map all uses of BC, but to examine how BC becomes textile material.

The two authors' experiments and the external cases are treated as different but complementary forms of material evidence. The experiments provide situated material knowledge from hands-on work with BC. The external cases extend the analysis across a wider range of textile material forms and production conditions. Together, they make it possible to compare how BC is grown, handled, transformed, and brought into textile systems. RtD and biotinkering therefore describe the mode of the authors' two experiments (E1, E2) rather than the method of the analysis as a whole, most of the sixteen cases are external works that the study analyses rather than conducts. This distinction matters because the paper's contribution is comparative and design-analytical: it maps where textile

interventions enter BC materialisation across practices, and uses the two experiments as first-hand probes into process stages: wet handling, drying, binding, and extrusion, that are often underreported in published cases.

3.2 Design-led biofabrication and biotinkering

The two authors' experiments were conducted as exploratory material probes through biotinkering. The aim was not to validate a composite or a filament, but to generate design knowledge through iterative, reflective making in which material behaviour is observed, responded to, and re-worked (Schön, 1983; Karana et al., 2015, 2023). In this mode, process notes, photographs, and retained material samples are treated not as illustration but as the primary evidence of what was learned. BC for both experiments was cultivated through static fermentation in laboratory-scale vessels (*Komagataeibacter* sp.). Each experiment was carried out over two iterations, with material parameters adjusted between rounds in response to observed behaviour.

Experiment E1- wet BC as a binder for textile waste (Figure 1). Wet, gel-state BC was combined with shredded pre-consumer wool waste, spread into a surface, and left to air-dry. Across the wet-to-dry transition, the key observations were: in the wet state, BC acted as a binder that held the fibres together and distributed them across a continuous surface. On drying, its stiffness, adhesion, and surface continuity changed markedly, such as stiffening, edge curling, cracking, delamination, shrinkage. The design knowledge generated is that the wet-to-dry transition is not a technical detail but the moment where the textile character of the biocomposite is decided. BC here does not substitute for a textile but co-forms a combined material with the waste it holds.

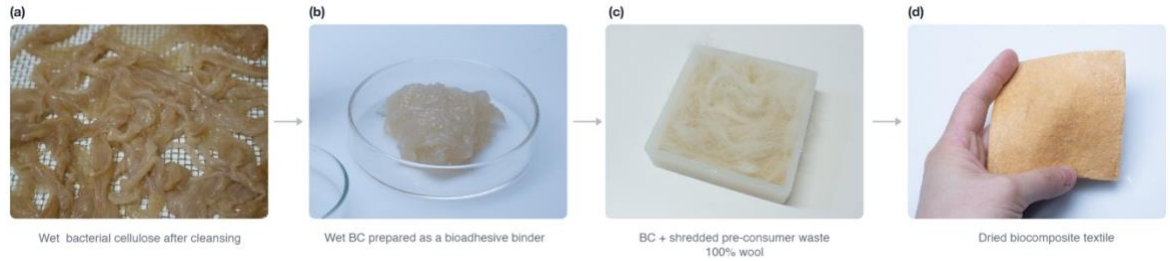
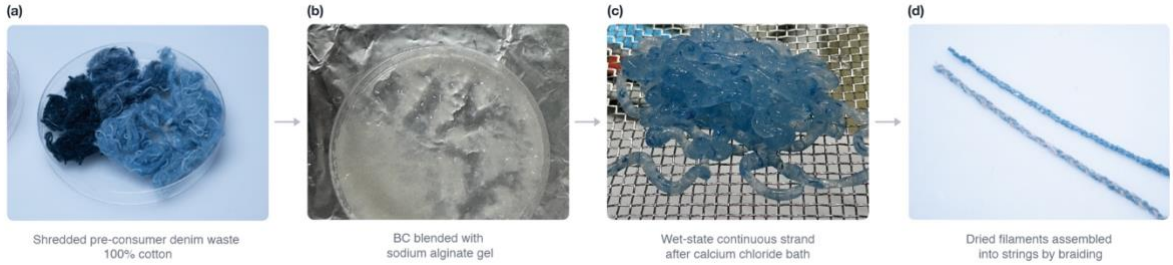


Figure 1. Experiment E1: wet BC used as a bioadhesive binder for shredded pre-consumer wool waste, across the wet-to-dry transition. (a) wet BC after cleansing; (b) BC developed into a glue-like binder; (c) BC combined with shredded wool waste; (d) the dried biocomposite textile.

Experiment E2- BC–alginate extrusion into filament *(Figure 2).* BC was blended with alginate gel, and extruded through a syringe into a 10% w/v calcium chloride solution, producing a wet-state



continuous strand. The observations were: the strand gelled and held form while wet and could be handled, on drying it developed discontinuity and breakage (Parisi et al., 2017). Although E2 contains denim fibres and therefore has a composite material composition, it is assigned to the filament trajectory because its primary materialisation logic is the extrusion and formation of a linear strand rather than the production of a sheet-like biocomposite. The design knowledge generated is that strand formation is not completed at the moment of extrusion but continues through gelation, handling, and drying, so continuity, shrinkage, and drying behaviour, not extrusion alone, govern whether BC becomes a filament.

Figure 2. Experiment E2: extrusion of a BC–alginate mixture into filament and its assembly into strings. (a) Shredded pre-consumer denim waste; (b) BC blended with sodium alginate gel; (c) wet-state continuous strand after a calcium chloride bath; (d) dried filaments assembled into strings by braiding.

3.3 Documented case selection and trajectory development

The fourteen external cases were selected from peer-reviewed articles, conference papers, design projects, material innovation reports, company documentation, and media sources. Searches combined terms such as “bacterial cellulose textile”, “bacterial cellulose fashion”, “kombucha textile”, “bacterial cellulose fibre”, “bacterial cellulose filament”, “bacterial cellulose yarn”, “bacterial cellulose coating”, “biofabricated textile”, and “microbial textile”. 53 potential cases were first identified.

Duplicates and speculative mentions were removed first. The final cases were retained when BC was a primary or materially significant component, when the case was directly related to textile material form, textile surface, fibre, filament, yarn, scaffold, or coating, and when enough process information was available to compare how BC was grown, processed, assembled, or applied.

Scientific papers and design or commercial cases were not treated as the same kind of evidence. Scientific papers were used mainly for technical and process information, while design and commercial cases were included when they showed textile materialisation practices not yet fully represented in academic literature. Cases such as Celium x Ganni and *Modern Synthesis* were therefore retained not as performance-validated evidence, but as documented design-oriented examples of sheet-based and scaffold-based BC textile materialisation.

The grouping started from familiar textile material forms: sheet, fibre, filament, yarn, composite, and coating. The final trajectories were shaped by the material practices observed in the cases. The focus was not only on final form, but on how BC was grown, processed, assembled with other materials, or applied to textile surfaces. Some cases could be read across more than one trajectory, for clarity, each was assigned to the trajectory that best described how BC became textile material in that case (see Table 1).

Filament and yarn are separated because they involve different ways of making strand-based textile material. In this paper, filament refers to strands formed through growth or extrusion, while yarn refers to strand formation through cutting, twisting, plying, or reassembling BC sheets.

Table 1. Bacterial cellulose cases included in the analysis, grouped by trajectory of textile materialisation

No.	Name	BC cultivation / production method	Processing techniques	Textile material form	References

E1	BC Biocomposite	Static fermentation (Komagataeibacter xylinus); harvested in wet, gel-like state	Wet-state binding of fragmented textile-waste fibres; air drying; observation of adhesion, stiffness, and surface continuity across wet-to-dry transition	Biocomposite textile	Experiment by author
E2	BC–alginate filament	Static fermentation (Komagataeibacter xylinus); BC blended with alginate gel and 1 g of shredded pre-consumer denim waste.	Syringe extrusion into calcium chloride bath; observation of wet-state continuity and drying-induced discontinuity	Filament	Experiment by author
C1	BioCouture	Static fermentation	Harvesting, washing, drying, cutting, stitching, shaping	Sheet	https://www.launch.org/innovators/suzanne-lee/
C2	Celium x Ganni blazer	Industrial microbial fermentation of bacterial cellulose	Harvesting, tanning/finishing, garment construction	Sheet	Polybion x GANNI
C3	Skin-friendly BC Textile	Static fermentation	Purification, plasticisation, drying, plant dyeing, reactive dyeing, digital printing, laser cutting	Sheet	https://doi.org/10.1007/978-3-032-08163-6_2
C4	Self-pigmenting BC textile	Static fermentation	Pigmentation during cultivation, harvesting, prototype assembly	Sheet	https://doi.org/10.1038/s41587-024-02194-3
C5	Tailor-shaped BC sheet	Static fermentation	Contacting surface-blocking cultivation; panel-shaped cultivation; shape-directed growth; reduced cutting	Sheet	https://doi.org/10.1177/0887302X17737177
C6	CelluLoop	Static fermentation from fruit waste, then dissolved for fibre spinning	dissolved for fibre spinning	Fibre	CelluLoop
C7	BCNF Macrofibres	Agitated fermentation	Jet-wet spinning, stretching methods,	Fibre	https://doi.org/10.3390/ijms25010069
C8	Nullarbor (Nanollose)	Industrial fermentation of food and agricultural waste (microbial cellulose)	Pulped and dissolved, blended with wood pulp, wet-spun into lyocell fibre on existing industrial lines	Fibre	https://nanollose.com/
C9	Bio-produced BC filaments	Direct fermentation designed to grow BC	Co-design with biological characteristics, controlled fermentation form-making	Filament	https://doi.org/10.3390/ma16144893

		into filament-like forms			
C10	Kami-ito yarns	Static fermentation in a rectangular mold	Sheet harvesting, strip cutting, twisting / paper-yarn logic	Yarn	https://doi.org/10.1016/j.ijbiomac.2025.141861
C11	BC Single Ply Yarns	Static fermentation	Cutting into narrow strips, twisting, drying	Yarn	https://doi.org/10.31274/itaa.8789
C12	Modern Synthesis microbial textile scaffold	Static fermentation within textile-like scaffold	In-situ growth, scaffold integration, material finishing	Biocomposite textile	Dezeen article
C13	Fermenting knits	Static fermentation	Harvesting and washing; blending into BC dope; kneading into knitted samples; short reincubation and fibre reassembly	Biocomposite textile	https://doi.org/10.1017/btd.2024.3
C14	Regenerated BC-coated cotton	Static fermentation from Kombucha, purified	Dissolved in DMAc/LiCl, applied to plain-woven cotton at four coating levels, regenerated on the fabric surface	Textile coating	https://doi.org/10.1080/15440478.2021.1958416

4. Findings: Six trajectories through which BC becomes textile material

The sixteen cases show that BC does not become textile through a single material route. It moves through six trajectories: sheet, fibre, filament, yarn, biocomposite textile, and textile coating, which are not fixed categories but varied approaches to integrating BC into textile systems, each shaped by the interplay between biological growth and material transformation. As summarised in Table 2, the trajectories describe different ways in which biological growth, processing, textile assembly, and surface treatment shape BC as a textile material.

Table 2. Six trajectories through which BC becomes textile material.

Textile form configuration	Related cases	Core Logic	Cultivation–processing relation
Sheet	C1, C2, C3, C4, C5	BC grows as a sheet-like pellicle and becomes textile through finishing, shaping, surface treatment, or cultivation-directed form.	Cultivation produces the base sheet; processing defines textile qualities. Colour or shape can also be formed during cultivation.
Fibre	C6, C7, C8	BC is treated as cellulose feedstock and transformed into fibre through dissolution, spinning, or stretching.	Cultivation acts as an upstream material production stage; textile formation happens mainly through post-processing.

Filament	C9, E2	BC is directed toward continuous linear form through controlled biological growth or post-harvest extrusion.	Cultivation and filament formation are integrated, with linear form emerging during biological growth (C9); Cultivation produces the BC feedstock in a BC–alginate–denim formulation, with formation occurring through extrusion, gelation, handling, and drying (E2).
Yarn	C10, C11	BC sheets are cut, twisted, and reassembled into yarn-like structures.	Cultivation produces the sheet first; post-processing translates it into a linear textile form.
Biocomposite textile	C12, C13, E1	BC forms with, on, or around textile structures, creating hybrid material systems.	Textile structure participates in formation, either by guiding growth or by being integrated after harvesting.
Textile coating	C14	BC is applied as a surface layer on an existing textile rather than forming the textile body; dissolved and regenerated onto the fabric, it modifies surface and barrier properties.	Cultivation and surface application are separated; post-processing enables localized surface treatment.

The six trajectories also produce visibly different textile material forms. *Figure 3* presents one representative case from each trajectory, showing how BC may remain sheet-like, become a fibre, filament, or yarn, form a biocomposite textile, or modify an existing textile through coating. The images are illustrative examples of the trajectory logic rather than equivalent forms of technical or performance evidence.

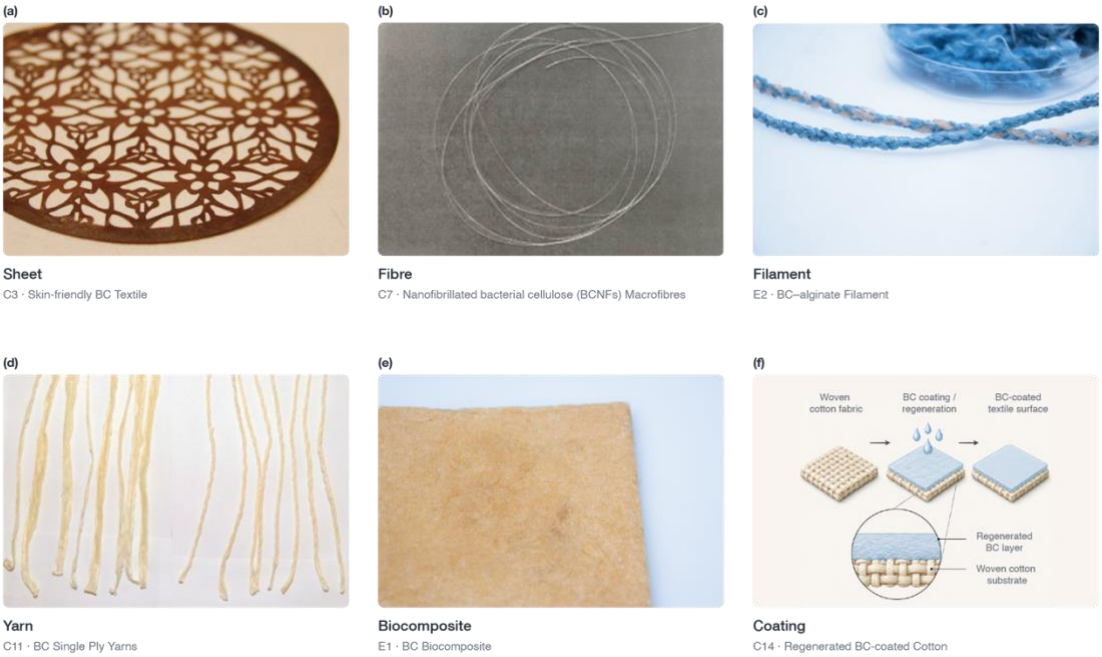


Figure 3. Representative material forms of the six bacterial cellulose textile materialisation trajectories: (a) sheet—skin-friendly BC textile (C3); (b) fibre—BCNF macrofibres (C7); (c) filament—the authors' BC–alginate filament (E2); (d) yarn—BC single-ply yarns (C11); (e) biocomposite textile—the authors' BC biocomposite (E1); and (f) textile coating—regenerated BC-coated cotton fabric (C14). Panels (c), and (e) show samples produced and photographed by the authors. Panel (f) is an AI-assisted schematic prepared under the authors' direction, based on the process reported by Kılınç et al. (2022) and does not depict a sample produced by the authors. Panel (a) is reproduced from Yu et al. (2026); Panel (b) is reproduced from Wang et al. (2024) under CC BY 4.0; panel (d) is reproduced from Harmon (2019) under CC BY 4.0.

4.1 Textile material forms: sheets, strands, composites, and coated surfaces

The sheet trajectory is the most established route for BC in textile and fashion contexts. In this trajectory, BC remains close to its pellicle-grown state and becomes textile through harvesting, washing, drying, finishing, shaping, cutting, stitching, dyeing, or surface treatment. BioCouture (C1), introduced above, together with Celium x Ganni (C2) and skin-friendly BC textile work (C3), follows this broad logic (Ng & Wang, 2016; Yu et al., 2026): BC is first grown as a sheet, and textile qualities are then produced through post-harvest treatment and garment or surface work.

Not all sheet cases follow a grow-first, shape-later sequence. Self-pigmenting BC textile (C4) and tailor-shaped BC textile (C5) move textile interventions into cultivation (Walker et al., 2025; Chan et al., 2018). In C4, colour formation is linked to the growth process. In C5, the growing environment is shaped so that BC forms closer to a textile panel. These cases show that sheet-based BC is not limited to post-processing.

The fibre, filament, and yarn trajectories move BC toward strand-based textile material, but they do so through different material practices. In the fibre trajectory, BC is treated as a cellulose source and transformed through dissolution, spinning, or stretching. Cases such as CelluLoop (C6) and BCNF (Reinforced cellulose composite macrofibres) (C7) place textile formation mainly after cultivation. Work on BC and regenerated cellulose shows how BC supports fibre production and textile-waste upcycling (Silva et al., 2024; Moriam et al., 2026). BC's biological growth produces the cellulose feedstock, but the textile form is produced through post-harvest processing.

At industrial scale, the same fibre trajectory appears in Nanollose's Nullarbor (C8), where microbial cellulose grown from food and agricultural waste is pulped, dissolved, and spun into lyocell fibre on existing industrial lines, blended with wood pulp. C8 shows that the fibre trajectory is not confined to laboratory or research settings: because BC here enters production as a cellulose feedstock rather than a finished sheet, it can be processed on established man-made-cellulosic infrastructure, the point at which textile interventions enter is also the point at which existing manufacturing can be reused.

The filament trajectory locates strand formation closer to growth or extrusion. Bio-produced BC filaments (C9) show how growth conditions can be designed to guide BC into filament-like forms (Morrow et al., 2023). The authors' BC–alginate–denim-waste extrusion experiment (E2) also belongs to this trajectory, but follows a post-harvest composite extrusion route. Here, cultivated BC is combined with alginate and shredded denim waste, extruded into a linear wet-state strand, and ionically gelled before drying. Although the formulation is materially composite, its dominant materialisation logic is filament formation. The experiment also shows that producing a strand at the point of extrusion does not guarantee continuity after drying.

The yarn trajectory is distinct from filament because it depends on textile construction after sheet formation. Kami-ito yarns (C10) and BC single-ply yarns (C11) form yarn-like structures from BC sheets

through cutting, twisting, drying, or paper-yarn logic (Takagi et al., 2025; Harmon, 2019). The textile question is not how to grow a strand directly, but how to rework a grown sheet into yarn-like material.

The biocomposite textile trajectory shows how BC becomes textile through material joining with textile waste, fibres, knitted structures, or scaffolds. Modern Synthesis (C12) grows BC within a textile-like scaffold, so the scaffold helps guide where the material settles. Fermenting Knits (C13) integrates broken-down BC fibres with knitted structures through blending, kneading, reincubation, fibre reassembly, washing, and drying (De Lara, 2024).

The authors' first experiment (E1) extends this trajectory by using wet BC as a binder for textile waste. Related textile-biofabrication studies also show how textile structures can shape biological material formation (Kaiser et al., 2024; Nerlich et al., 2023). In these cases, BC does not simply replace a textile material. It works with existing textile matter to form a combined material. The textile coating trajectory positions BC as surface work rather than as the main textile body.

In regenerated BC-coated cotton (C14), purified kombucha-derived BC is dissolved in DMAc/LiCl, applied to plain-woven cotton, and regenerated on the textile surface through coagulation baths (Kılınç et al., 2022). The regenerated BC progressively covers the cotton yarns and fills the fabric pores, increasing thickness, thermal conductivity, water-vapour permeability, and thermal stability, while reducing air permeability and tensile strength. Here, the textile surface remains central: BC modifies an existing textile substrate rather than becoming the whole textile material. Work on BC adhesion to treated cotton also shows the relevance of surface treatment for all-cellulose textile materials (Ogrizek et al., 2024). This trajectory is important because it shows that BC's textile relevance can also lie in coating, surface treatment, and localized material application.

4.2 Where textile interventions enter the design-led biofabrication and biotinkering process

Across the six trajectories, textile interventions enter the material process at different points. The comparison shows that BC does not become textile at a single point in the process. It becomes textile through growth, wet handling, spin or extrude, fibre or strand making, textile assembly, and surface treatment. This is the main design finding of the study: BC's textile potential is distributed across material work *(Figure 4)*, rather than located only in the final object.

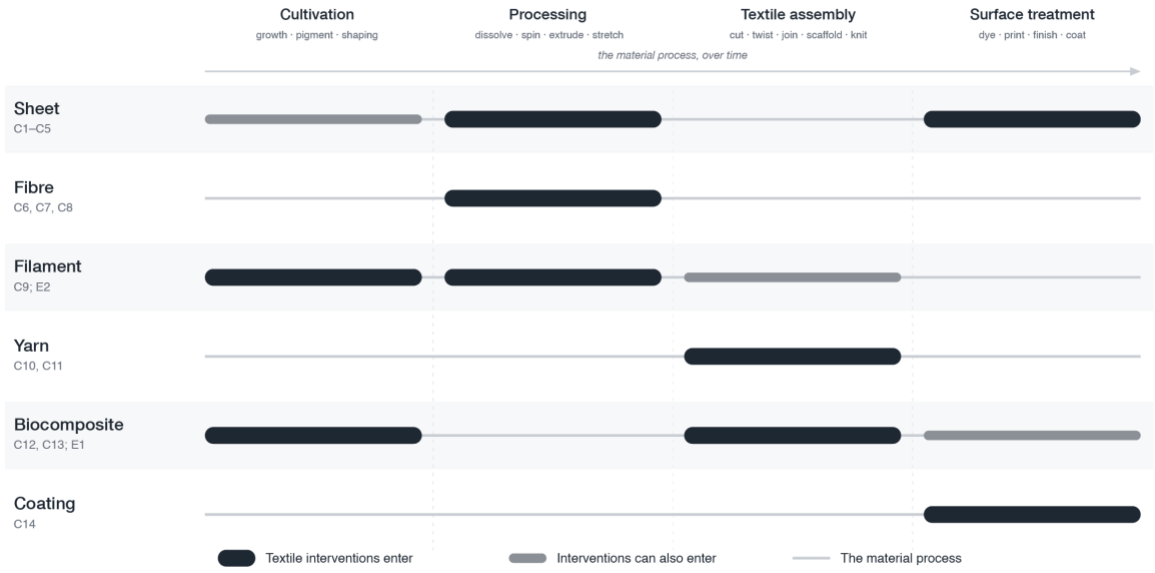


Figure 4. Textile interventions across the six trajectories of BC materialisation. Bars mark where textile interventions enter the material process for each trajectory (dark = textile interventions enter; grey = intervention can also enter; thin line = the material process, with BC present throughout).

5. Discussion

5.1 From substitute material to textile materialisation

The findings shift the discussion from what BC can replace to how BC is grown, processed, and treated within textile systems. When BC is treated mainly as a leather substitute, its textile role is tied to the sheet. This makes the material easier to compare with existing materials, but it also reduces the design space. The six trajectories show that BC may become textile as a sheet, fibre, filament, yarn, biocomposite textile, or textile coating.

This distinction matters for textile design research. A substitute-material frame asks whether BC can integrate or improve an existing material in the textile industry. A textile materialisation frame asks where textile formation enters the material fabrication process. In design research on living materials, material development is understood as part of design practice rather than only technical preparation (Camere & Karana, 2018; Karana et al., 2023). BC should therefore not be described only as a final material outcome. It should also be understood through the material practices that produce its textile qualities. These practices include growing, harvesting, washing, drying, cutting, extruding, spinning, twisting, binding, coating, and working with textile substrates.

Mapping the cases over time *(Figure 5)* offers tentative support for this reframing: the sheet trajectory spans the longest period in the selected cases. Most non-sheet cases cluster from 2023 onwards, although earlier fibre, yarn, and coating precedents are also present.

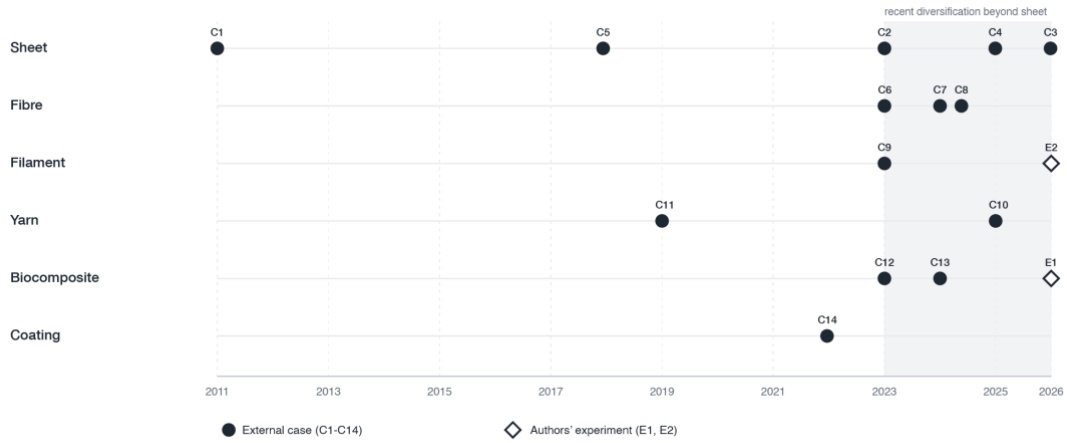


Figure 5. The sixteen cases mapped by year (2011–2026) and materialisation trajectory, showing that diversification beyond the sheet trajectory is recent.

5.2 Textile interventions across design practice of growth, processing, and assembly

The study also clarifies the role of Material-Driven Design and Material-Driven Textile Design. Material-Driven Design helps frame BC as a material that can guide design inquiry through its behaviour (Karana et al., 2023).

The authors' experiments make this point visible: E1 shows the wet-to-dry transition as a design concern, and E2 shows that strand formation continues through gelation, handling, and drying, textile knowledge enters through material behaviour rather than final form.

Biotinkering is therefore not only a way of making samples. In this study, it helps reveal material behaviour that is difficult to see from finished artifacts or documented case studies alone. It keeps attention close to wetness, fragility, shrinkage, adhesion, and the unstable passage from biological growth to textile material form. This links the two authors' experiments to material tinkering as a way of learning through direct material engagement (Parisi et al., 2017).

5.3 Material state, textile function, and limits

A further issue concerns the state of BC after cultivation. Being produced by bacteria does not mean that the resulting textile material remains living or biologically active. It is more precise to place BC textiles along a *gradient of aliveness*, defined by how processing eliminates, maintains, or promotes the organisms involved in production.

At one end, purification, drying, heating, or chemical treatment renders BC an inert cellulose substrate in which the bacteria are no longer active. In the middle, BC may act as a carrier for separately added functional agents, or as a surface layer whose behaviour derives from its structure rather than from living cells. At the other end, BC can be kept hydrated and unpurified, or deliberately combined with other organisms, as part of a biologically active system.

These are different material states produced by different treatments, and should not be conflated. Claims such as antimicrobial behaviour, responsiveness, or bioactivity therefore need to be tied to a specific position on this gradient, to a named material state and mechanism rather than inferred from BC's microbial origin. This distinction is particularly important for skin-facing textile materials, where comfort, care, skin wellness, and health-related claims carry different evidentiary requirements and should not be treated as equivalent (Yu, Duarte Poblete, & Rognoli, 2026). Locating a BC textile on the aliveness gradient clarifies whether its function is grown, retained, or added, and what evidence is required, alongside textile concerns such as wettability, comfort, durability, and dimensional stability (Dip et al., 2026; Provin et al., 2021; Wood et al., 2023; Wood et al., 2026).

The study also has limits. The six trajectories are not a complete taxonomy of BC textile applications. They are based on selected cases, available documentation, and two exploratory material probes. The contribution of the paper is therefore not a definitive classification, but a clearer account of how BC becomes textile material through growth, processing, textile assembly, and surface treatment.

The study also does not address BC from an industrial manufacturing perspective. Most of the cases examined are design-led, exploratory, or small-scale, and few engage questions of scalability, throughput, or process control; the trajectories therefore describe how BC becomes textile in design and research practice rather than how it might be manufactured at scale. Assessing the manufacturing opportunities and limitations of each trajectory, and whether wet-state and drying behaviours can be controlled industrially, remains a direction for future work.

The Nullarbor case (C8) is the main exception, indicating that at least the fibre trajectory can align with existing man-made-cellulosic manufacturing; the manufacturability of the other trajectories remains largely untested.

Reading BC through materialisation trajectories also reframes where sustainability questions should be asked. The trajectories distribute material and processing burdens differently: the sheet trajectory keeps BC close to its grown state with minimal processing, whereas the fibre trajectory adds dissolution and spinning, and the extrusion route in the filament trajectory introduces alginate and a calcium chloride bath. A process-based account therefore suggests that environmental impacts cannot be inferred from BC's material identity alone, they depend on feedstock, cultivation, purification, transformation, use, and end-of-life conditions.

6. Conclusion

This study has examined the materialisation trajectories of bacterial cellulose (BC) as a textile material. Across sixteen cases, comprising two authors' experiments and fourteen external design and interdisciplinary cases, it identified six trajectories: sheet, fibre, filament, yarn, biocomposite textile, and textile coating.

The findings show that BC's textile potential is not located in one stable form. It emerges through growth, handling, processing, textile assembly, and surface treatment. This shifts the discussion away from BC as primarily a leather substitute and toward how BC is made textile through material practice.

The significance of this account extends beyond BC. First, the trajectories offer a transferable analytical lens for other biofabricated and grown materials, such as mycelium, algae, and protein-based materials, that are also frequently narrowed to a single substitute form; asking where textile interventions enter their materialisation can open comparable ranges of textile possibility, particularly for materials whose trajectories remain underexplored. Second, by locating textile formation at specific process stages, the framework reframes where questions of performance and sustainability should be posed: as properties of a trajectory rather than of the material as such. Third, the paper shows how biotinkering can support Material-Driven Design by making visible the wet-state behaviour, drying instability, binding, and extrusion that finished cases rarely document, an approach that other researchers may use to surface process knowledge in early-stage biomaterials.

These trajectories remain open and non-exhaustive. Further research should examine the performance, durability, comfort, scalability, and functional mechanisms of BC-based textile materials, especially for skin-facing and coated applications.

References

Absharina, D., Padri, M., Veres, C., & Vágvölgyi, C. (2025). Bacterial cellulose: From biofabrication to applications in sustainable fashion and vegan leather. *Fermentation*, *11*(1), 23. https://doi.org/10.3390/fermentation11010023

Absharina, D., Putra, F. J. N., Ogino, C., Kocsubé, S., Veres, C., & Vágvölgyi, C. (2025). Bacterial cellulose production in co-culture systems: Opportunities, challenges, and future directions. *Applied Microbiology*, *5*(3), 92. https://doi.org/10.3390/applmicrobiol5030092

Avci, T., Çalhan, A., Ayar-Sumer, E. N., Dogan, C., Kaddami, H., Gunduz, O., & Narayan, R. (2026). Sustainable and eco-friendly bacterial cellulose production using agro-forest waste for food packaging applications: A review. *Sustainable Food Technology*, Advance online publication. https://doi.org/10.1039/D5FB00923E

Bell, F., Alistar, M., & Buechley, L. (2026). Integrating biomaterials and interactive technologies: Practice-based perspectives on the growth of biodesign within human-computer interaction. *Biotechnology Design*, *4*, e3. https://doi.org/10.1017/S2977905725100036

Biranje, S. S., Kaushik, S., Marewad, D., Yadav, A., Vankundre, V., Panse, M., Joshi, I., Goli, A., Shahid, M., Kulkarni, K., Liu, J., & Adivarekar, R. (2024). Applications of nanocellulose and its derivatives in developing sustainable textiles. *Cellulose*, *31*(9), 5343–5379. https://doi.org/10.1007/s10570-024-05935-4

Buso, A., McQuillan, H., Voorwinden, M.,and Karana, E.(2023) Towards Performative Woven Textile-form Interfaces, in Tincuta Heinzel, Delia Dumitrescu, Oscar Tomico, Sara Robertson (eds.), *Proceedings of Textile Intersections Conference 2023*, 20 - 23 September, London, United Kingdom. https://doi.org/10.21606/TI-2023/106

Camere, S., & Karana, E. (2018). Fabricating materials from living organisms: An emerging design practice. *Journal of Cleaner Production*, *186*, 570–584. https://doi.org/10.1016/j.jclepro.2018.03.081

Chan, C. K., Shin, J., & Jiang, S. X. K. (2018). Development of tailor-shaped bacterial cellulose textile cultivation techniques for zero-waste design. *Clothing and Textiles Research Journal, 36*(1), 33–44. https://doi.org/10.1177/0887302X17737177

Cleveland, D., Nayak, R., & Joseph, F. (2024). Designing with bacterial cellulose: A pilot study using localized food sources to grow innovative materials in Vietnam. *Journal of Textile Design Research and Practice*, *12*(3), 192–222. https://doi.org/10.1080/20511787.2025.2472528

Cohen, N., Sicher, E., Ayala-Garcia, C., Merino Sanchez-Fayos, I., Conterno, L., & Ugur Yavuz, S. (2023). Innocell bioreactor: An open-source development to produce biomaterials for food and packaging based on fermentation processes. *Fermentation, 9*(10), 915. https://doi.org/10.3390/fermentation9100915

Collet, C. (2018). Biotextiles: Evolving Textile DesignPractices for the Bioeconomy and the EmergingOrganism Industry. In: N., Nimkulrat, U.,Raebuild, & A., Piper eds. Soft Landing. *Cumulus*, Aalto University, pp. 87–98.

Collet, C.(2017) Grow-made Textiles, in Karana, E., Giaccardi, E., Nimkulrat, N., Niedderer, K., Camere, S. (eds.), *EKSIG 2017: Alive. Active. Adaptive.*, 19-20 June 2017, Delft and Rotterdam, The Netherlands. https://doi.org/10.21606/eksig2017.101

De Lara, C. (2024). Fermenting knits: A material-driven exploration of knit-based bacterial cellulose biocomposite textile materials through fibre reassembly. *Research Directions: Biotechnology Design, 2*, e3. https://doi.org/10.1017/btd.2024.3

Dip, T. M., Rahman, M., & Wood, J. (2026). Fabrication and characterization of electrospun sodium alginate–bacterial cellulose nanocomposite scaffolds for potential wound dressing application. *Macromolecular Materials and Engineering*, *311*(1), e00232. https://doi.org/10.1002/mame.202500232

D'Itria, E., Guarino, N., Papile, F., & Yu, Z. (2026). Closing the loop: A pilot study on circular feedstock in the fashion industry. *Diid*, *digital special issue 3,* 204–221. https://doi.org/10.30682/diidmics25o

Duarte Poblete, S. S., Anselmi, L., and Rognoli, V. (2024) Materials designers and the translational approach: a case from a product design company, in Gray, C., Ciliotta Chehade, E., Hekkert, P., Forlano, L., Ciuccarelli, P., Lloyd, P. (eds.), *DRS2024: Boston*, 23–28 June, Boston, USA. https://doi.org/10.21606/drs.2024.1080

Duarte Poblete, S. S., Anselmi, L., & Rognoli, V. (2023). Emerging materials fostering interdisciplinary collaboration in materials design. In *Proceedings of the 14th International Conference on Applied Human Factors and Ergonomics*. https://doi.org/10.54941/ahfe1002978

Fernandes, A. M., Silva, T. L., & Silva, C. J. (2025). Bacterial cellulose: A new biomaterial for the textile industry. *Next Sustainability*, *6*, 100211. https://doi.org/10.1016/j.nxsust.2025.100211

Gao, C., Liu, Y., Gu, Z., Li, J., Sun, Y., Li, W., Liu, K., Xu, D., Yu, B., & Xu, W. (2024). Hierarchical structured fabrics with enhanced pressure sensing performance based on orientated growth of functional bacterial cellulose. *Advanced Fiber Materials*, *6*(5), 1554–1568. https://doi.org/10.1007/s42765-024-00435-3

Harmon, J., (2019) "Cotton Versus Bacterial Cellulose: A Comparison of Single Ply Yarns*", International Textile and Apparel Association Annual Conference Proceedings 76(1)*. https://doi.org/10.31274/itaa.8789

Kaiser, R., Dade-Robertson, M., Elsacker, E., & Scott, J. (2024). Knit-mycelium hybrids: Textile design strategies for biofabrication systems based on mycocrete injection in 3D knitted tubular formworks. *Journal of Textile Design Research and Practice*, *12*(1–2), 10–35. https://doi.org/10.1080/20511787.2024.2389613

Kapsali, V., & Hall, C. A. (2023). Bioinspired textiles; observations from the evaluation of a novel practice-based framework linking lessons on sustainable/circular design from biology to textile practice. *Journal of Textile Design Research and Practice*, *11*(1–2), 148–167. https://doi.org/10.1080/20511787.2023.2240062

Kapsali, V., Silva, E. S., & Hall, C. A. (2024). Biology-informed textile design: A mixed-method study of textile professionals in the UK. *Journal of Textile Design Research and Practice*, *12*(3), 223–247. https://doi.org/10.1080/20511787.2025.2523620

Karana, E., Barati, B., & Rognoli, V. (2015). Material Driven Design (MDD): A Method to Design for Material Experiences. 9(2), 20. *International Journal of Design*, 9(2), 35-54.

Karana, E., McQuillan, H., Rognoli, V., & Giaccardi, E. (2023). Living artefacts for regenerative ecologies. *Research Directions: Biotechnology Design, 1*, e16. https://doi.org/10.1017/btd.2023.10

Kılınç, M., Ay, E., & Kut, D. (2022). Thermal, chemical and mechanical properties of regenerated bacterial cellulose coated cotton fabric. *Journal of Natural Fibers, 19*(14), 7834–7851. https://doi.org/10.1080/15440478.2021.1958416

Lazaro Vasquez, E. S., Arduini, V., Sandry, E. W., Houser, K., Golla, S., & Alistar, M. (2026). Bioactuated tapestry: An installation exploring temporalities of textile craft and moisture-responsive biomaterials. In *Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction*, 1–4. https://doi.org/10.1145/3731459.3779152

Mihaleva, G. (2021). Bio matter in creative practises for fashion and design. *AI & SOCIETY*, *36*(4), 1361–1365. https://doi.org/10.1007/s00146-020-00957-5

Miranda, K. W. E., Sá, G. V., Frota, M. M., Dos Santos Neto, J. P., De Carvalho, M. Â. A. P., Nunes, N. M. V., Machado, M. T. D. C., De Souza, T. M., Bastos, M. D. S. R., & De Vasconcelos, L. B. (2026). Valorization of

amazonian fruit agro-industrial by-products for sustainable production of bacterial cellulose nanofibres for food bio-packaging. *Polymer Bulletin*, *83*(6), 290. https://doi.org/10.1007/s00289-026-06348-y

Morgan, L., Mosse, A., Dragosavac, M., & Kane, F. (2024). Convenors' note – biotextiles and sustainable textiles. *Journal of Textile Design Research and Practice*, *12*(1–2), 64–69. https://doi.org/10.1080/20511787.2024.2461910

Moriam, K., Owens, C. E., Kroo, L., Ghann, W., Uddin, J., Pitkänen, L., Hummel, M., & McKinley, G. H. (2026). Enhancing the spinnability of cellulose-based textile waste by doping with high molecular weight bacterial cellulose. *Biomacromolecules*, *27*(4), 2612–2628. https://doi.org/10.1021/acs.biomac.5c02370

Morrow, R., Ribul, M., Eastmond, H., Lanot, A., & Baurley, S. (2023). Bio-producing bacterial cellulose filaments through co-designing with biological characteristics. *Materials*, *16*(14), 4893. https://doi.org/10.3390/ma16144893

Moser, J., Fleck, K.,and Posch, I.(2023) Growing Patterns: Textile Design with Living Pigments, in Tincuta Heinzel, Delia Dumitrescu, Oscar Tomico, Sara Robertson (eds.), *Proceedings of Textile Intersections Conference 2023*, 20 - 23 September, London, United Kingdom. https://doi.org/10.21606/TI-2023/109

Nayak, R., Cleveland, D., Tran, G., & Joseph, F. (2024). Potential of bacterial cellulose for sustainable fashion and textile applications: A review. *Journal of Materials Science*, *59*(16), 6685–6710. https://doi.org/10.1007/s10853-024-09577-6

Nerlich, P., Sherry, A., Deary, M., Morrow, R.,and Scott, J.(2023) Sensory textile-bacterial hybrids: textile-bacteria fusion to impart forest-associated scents, in Tincuta Heinzel, Delia Dumitrescu, Oscar Tomico, Sara Robertson (eds.), *Proceedings of Textile Intersections Conference 2023*, 20 - 23 September, London, United Kingdom. https://doi.org/10.21606/TI-2023/111

Ng, F. M. C., & Wang, P. W. (2016). Natural self-grown fashion from bacterial cellulose: A paradigm shift design approach in fashion creation. *The Design Journal*, *19*(6), 837–855. https://doi.org/10.1080/14606925.2016.1208388

Obregón, C. (2026). Biodesign rethinking fashion through biology and technology. *Journal of Textile Engineering & Fashion Technology*, *12(1):11–16*. https://doi.org/10.15406/jteft.2026.12.00443

Ogrizek, L., Lamovšek, J., Primc, G., Leskovšek, M., Vesel, A., Mozetič, M., & Gorjanc, M. (2024). Improved adhesion of bacterial cellulose on plasma-treated cotton fabric for development of all-cellulose biocomposites. *Molecules*, *29*(21), 5009. https://doi.org/10.3390/molecules29215009

Parisi, S., Rognoli, V., & Sonneveld, M. (2017). Material tinkering. An inspirational approach for experiential learning and envisioning in product design education. *The Design Journal*, *20*(sup1), S1167–S1184. https://doi.org/10.1080/14606925.2017.1353059

Phan, H. N., & Okubayashi, S. (2025). Modification of dehydrated bacterial cellulose with glycerol and succinic acid by using padding method for textile applications. *Research Journal of Textile and Apparel*, *29*(2), 371–388. https://doi.org/10.1108/RJTA-10-2023-0111

Provin, A. P., Reis, V. O. D., Hilesheim, S. E., Bianchet, R. T., Dutra, A. R. D. A., & Cubas, A. L. V. (2021). *Use of bacterial cellulose in the textile industry and the wettability challenge—A review*. Cellulose 28, 8255–8274 (2021). https://doi.org/10.1007/s10570-021-04059-3

Ribul, M., Goldsworthy, K., & Collet, C. (2021). Material-driven textile design (MDTD): A methodology for designing circular material-driven fabrication and finishing processes in the materials science laboratory. *Sustainability*, *13*(3), 1268. https://doi.org/10.3390/su13031268

Roussel, V., Haute, L., Dhulster, P., & Teyssier, M. (2024). Processes, fabrication and design with kombucha bacterial cellulose: Mapping practices. In *Proceedings of the 28th International Symposium on Electronic Art*. https://doi.org/10.69564/ISEA2023-81-full-Roussel-et-al-Mapping-Practices

Schön, D. A. (1983). *The reflective practitioner: How professionals think in action*. Basic Books.

Silva, F. A. G. S., Schlapp-Hackl, I., Nygren, N., Heimala, S., Leinonen, A., Dourado, F., Gama, M., & Hummel, M. (2024). Upcycling of cellulosic textile waste with bacterial cellulose via Ioncell® technology. *International Journal of Biological Macromolecules*, *271*, 132194. https://doi.org/10.1016/j.ijbiomac.2024.132194

Sözcü, Ş., Wiener, J., Frajová, J., Venkataraman, M., Tomková, B., Kalmár, J., Forgács, A., & Militký, J. (2026). Effect of drying methods on *Acetobacter xylinum* bacterial cellulose aerogels and cryogels. *Scientific Reports*, *16*(1), 12264. https://doi.org/10.1038/s41598-026-42244-1

Stappers, P. and Giaccardi, E. (2014). Research through Design. IxDF - Interaction Design Foundation. https://ixdf.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/research-through-design

Takagi, A., Niu, X., Wang, P., Mehling, M., Pritchard, S., Hahn, S., Young, H., Guo, T., Lu, Y., & Rojas, O. J. (2025). High strength kami-ito yarns from microbial cellulose biofilms. *International Journal of Biological Macromolecules*, *307*, 141861. https://doi.org/10.1016/j.ijbiomac.2025.141861

Tang, Y., Xue, Y., Yuan, J., & Xu, J. (2025). Research and application of bacterial cellulose as a fashionable biomaterial in dyeing and printing. *Sustainability*, *17*(17), 7631. https://doi.org/10.3390/su17177631

Vasili, E., Azimi, B., Raut, M. P., Gregory, D. A., Mele, A., Liu, B., Römhild, K., Krieg, M., Claeyssens, F., Cinelli, P., Roy, I., Seggiani, M., & Danti, S. (2025). A green method for bacterial cellulose electrospinning using 1-butyl-3-methylimidazolium acetate and γ-valerolactone. *Polymers*, *17*(9), 1162. https://doi.org/10.3390/polym17091162

Verran, J., Wood, J., Redfern, J., Moravej, H., & Radclyffe-Thomas, N. (2023). Hands on biofilm! A multidisciplinary public engagement event using kombucha tea pellicle as an accessible example of biofilm. *Biofilm*, *6*, 100169. https://doi.org/10.1016/j.bioflm.2023.100169

Walker, K. T., Li, I. S., Keane, J., Goosens, V. J., Song, W., Lee, K.-Y., & Ellis, T. (2025). Self-pigmenting textiles grown from cellulose-producing bacteria with engineered tyrosinase expression. *Nature Biotechnology*, *43*(3), 345–354. https://doi.org/10.1038/s41587-024-02194-3

Wang, J., Tavakoli, J., & Tang, Y. (2019). Bacterial cellulose production, properties and applications with different culture methods – A review. *Carbohydrate Polymers*, *219*, 63–76. https://doi.org/10.1016/j.carbpol.2019.05.008

Wang, R., Fujie, T., Itaya, H., Wada, N., & Takahashi, K. (2024). Force-induced alignment of nanofibrillated bacterial cellulose for the enhancement of cellulose composite macrofibers. *International Journal of Molecular Sciences, 25*(1), 69. https://doi.org/10.3390/ijms25010069

Wood, J., Verran, J., Randviir, E., & Redfern, J. (2026). A new application of bacterial cellulose in textiles and fashion: Using Kombucha-derived biofilm to remove dye from polluted water. *Scientific Reports*, *16*(1), 9445. https://doi.org/10.1038/s41598-026-39271-3

Wood, J., Verran, J., & Redfern, J. (2023). Bacterial cellulose grown from kombucha: Assessment of textile performance properties using fashion apparel tests. *Textile Research Journal*, *93*(13–14), 3094–3108. https://doi.org/10.1177/00405175231152668

Yu, Z., Rognoli, V., & Bertola, P. (2026). Biodesign in textile innovation: Emerging applications of bacterial cellulose to fashion. In A. Mateus, G. Patrocinio, & S. Leonor (Eds.), *What's Around Design?* (pp. 21–32). Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-08163-6_2

Yu, Z., Rognoli, V., & Duarte Poblete, S. S. (in press). From Material Tinkering to Biotinkering: Infrastructuring a Biodesign Laboratory for Emerging Materials Practices. *Design Commit.*

Yu, Z., Duarte Poblete, S. S., & Rognoli, V. (2026). Framing bacterial cellulose at the skin interface: Comfort, care, and skin wellness in biodesign. *Biotechnology Design, 4*, e26. https://doi.org/10.1017/S2977905726100511

About the Authors:

Ziqian Yu, PhD candidate and manager of the Biodesign Lab at Politecnico di Milano. Her research investigates biofabricated materials, particularly bacterial cellulose, through Material Driven Design practice within the biodesign context, promoting skin wellness through textile innovation.

Valentina Rognoli, Associate Professor at Politecnico di Milano and founder of the Materials Design for Transition group and the Biodesign Lab. Her research explores material design, biodesign, and regenerative approaches, integrating bioregional perspectives, craft knowledge, and emerging materials across diverse design disciplines.

Acknowledgement: The first author acknowledges support from the China Scholarship Council (CSC). The authors thank the Biodesign Lab (B.LAB), Department of Design, Politecnico di Milano. Experiments E1 and E2 were conducted within the MICS (Made in Italy – Circular and Sustainable) Extended Partnership, funded by the European Union NextGenerationEU under the PNRR (Mission 4, Component 2, Investment 1.3; D.D. 1551, 11 October 2022; PE00000004). The views expressed are solely those of the authors and do not necessarily reflect those of the European Union or the European Commission. Generative AI was used solely to assist the visual rendering of Figure 3(f); the authors defined and verified its scientific content and take responsibility for the final figure.